

OMANSHU BHATT

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Portfolio: <https://omanshu-bhatt-portfolio.vercel.app/> Github: <https://github.com/Omanshu09> LinkedIn: <https://www.linkedin.com/in/omanshubhatt/>

PROFESSIONAL SUMMARY

Software Developer specializing in AI/ML and full-stack development. Builds end-to-end systems — from RAG-based retrieval and LLM reasoning pipelines to real-time collaborative applications. Comfortable shipping production-grade code across the stack.

TECHNICAL SKILLS

Languages: Python, Go, Java, JavaScript, C

Web & Frameworks: React.js, FastAPI, Node.js, Django, HTML, CSS, Typescript

Data & Databases: Elasticsearch, PostgreSQL, Redis, Neon

AI / ML: PyTorch, LangChain, Generative AI, RAG, Vector Databases

Cloud, DevOps & Systems: Docker, Kubernetes, Kafka, CI/CD (Jenkins, GitHub Actions)

Distributed & Realtime Systems: Yjs, CRDTs, Ably, IndexedDB, REST APIs, Distributed State Synchronization

Core CS: Data Structures & Algorithms, Operating Systems, Database Management Systems, Object-Oriented Programming

FEATURED PROJECTS

Web Research & Evidence Engine

- Built a research-driven web evidence search engine that performs query processing, multi-source web retrieval, content extraction, relevance filtering, ranking to identify high-value sources and aggregates retrieved evidence to detect agreement and contradictions before generating structured, source-backed responses through LLM-based evidence synthesis.
- Architected the end-to-end application with Python/FastAPI services for search, retrieval, ranking and evidence analysis, a React-based research interface and Dockerized deployment, integrating retrieval and processing into a reliable pipeline for automated research.
- Tech:** Web Search, Information Retrieval, Evidence Extraction, RAG, LLM, FastAPI, React, Docker
- Benchmarked across 100+ research queries, achieving ~85% relevant source retrieval with sub-10s median response time.

Code: <https://github.com/Omanshu09/Search-Engine>

Live: <https://search-engine-frontend-xi.vercel.app/>

Real-Time Collaborative Engineering Workspace

- Built a real-time collaborative engineering workspace enabling multiple users to edit a shared CRDT-based Notepad with near-instant synchronization, while maintaining an independent Python execution environment and shared execution output; designed the system to remain resilient during connectivity loss through automatic state recovery.
- Deployed full-stack system with a React/Vite frontend on Vercel, FastAPI/Node.js backend services on Render, Ably for low latency realtime messaging and synchronization, Yjs CRDTs with IndexedDB persistence for conflict free collaboration and offline durability, Neon PostgreSQL for persistent room state, and E2B isolated sandboxes for secure Python execution.
- Tech:** React, TypeScript, Vite, Node.js, Express, Yjs, CRDT, Ably, IndexedDB, PostgreSQL, Neon, E2B, REST APIs, Vercel, Render
- Instant collaborative synchronization during multi user editing, with offline local state retention and isolated Python execution

Code: <https://github.com/Omanshu09/Nexus-deployment>

Live: <https://nexus-deployment-frontend.vercel.app/>

EXPERIENCE

Trainee Intern | Indian Oil Corporation Limited (IOCL) | July 2025 – August 2025

- Worked with operational data and database systems to analyze trends, identify recurring patterns, and support data-backed improvements in engineering processes.
- Collaborated directly with IOCL engineers and technical teams, gaining hands-on experience in database management, enterprise engineering workflows, and quality-focused problem solving.

Virtual Intern — AI/ML & Robotics | Kodacy | July 2024 – August 2024

- Applied Python to AI/ML and data-analysis tasks, working with datasets to extract patterns, evaluate results, and build practical solutions.
- Developed Python-based robotics simulation prototypes, gaining exposure to algorithmic problem solving, automation and intelligent systems.

CERTIFICATIONS

- Oracle — Agentic AI Certified Foundations Associate
- TCS iON Career Edge - Young professional
- NASA — Hyperspectral Data for Land and Coastal Systems
- Deloitte — Cyber Job Simulation
- Commonwealth Bank — Software Engineering Job Simulation
- University of Oxford — AI, Justice and rule of law

EDUCATION

Bachelor of Technology (B.Tech) - Computer Engineering

Govind Ballabh Pant University of Agriculture and Technology, Pantnagar | 2022 – 2026 | CGPA: 7.2

Beersheba Senior Secondary School, Almora

Class XII | 2021 – 2022 | 90.4%

Class X | 2019 – 2020 | 95.3%